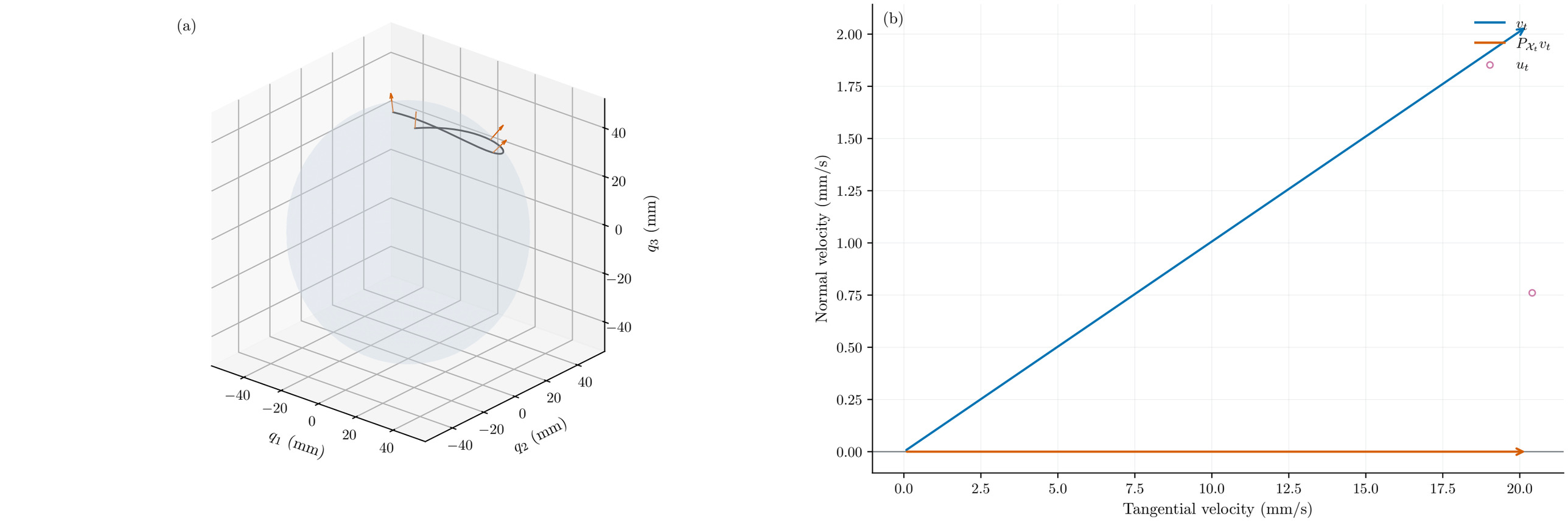

(a)
q_1 (mm)
q_2 (mm)
q_3 (mm)
−40
−20
0
20
40
(b)
v_t
$P_{\mathcal{X}_t} v_t$
u_t
Normal velocity (mm/s)
Tangential velocity (mm/s)
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0